COMPUTER Network

22CSE62 –M.5.2

BHASKAR S V

Module 5 -Application layer and Security mechanisms

- **Overview of:**
  - Domain Name Space,
  - Remote Logging, Email, FTP, WWW, HTTP
  - Security – Network Layer Security, Application Layer Security

Network Layer Security

- Network-layer security provides protection at the IP layer of the TCP/IP protocol suite.
- Security can be applied between:
  - Two hosts
  - Two routers
  - A host and a router
- It protects data while packets travel across the network infrastructure.

Key Objectives

- Authentication
- Confidentiality
- Integrity
- Secure communication across insecure networks

Purpose of Network-Layer Security

- Protect applications that directly use network-layer services.
- Secure routing protocols and packet forwarding mechanisms.
- Provide protection for UDP-based applications because UDP is connectionless and cannot use transport-layer security protocols such as TLS.

Introduction to IPSec

IP Security (IPSec)

- IPSec stands for **Internet Protocol Security.**
- Developed by the **Internet Engineering Task Force (IETF).**
- IPSec is a collection of protocols that provide security services at the IP layer.

Features of IPSec

- Packet authentication
- Data confidentiality through encryption
- Data integrity verification
- Protection against replay attacks

Introduction to IPSec

Why IPSec is Important

- Secures communication over public networks such as the Internet.
- Enables secure Virtual Private Networks (VPNs).
- Provides end-to-end security at the network layer.

IPSec Works By

- Adding security headers and trailers to IP packets.
- Encrypting or authenticating packet contents.
- Protecting communication transparently to applications.

IPSec Operating Modes

1. Transport Mode

- Protects only the transport-layer payload.
- Original IP header remains unchanged.
- Used mainly for host-to-host communication.

2. Tunnel Mode

- Protects the entire original IP packet.
- Adds a new IP header.
- Commonly used in VPNs and router-to-router communication.

IPSec Operating Modes

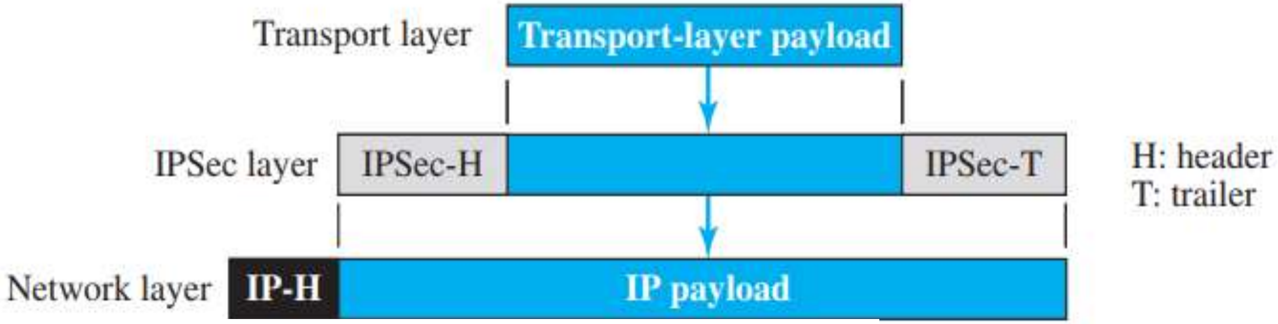
Major Difference

Feature	Transport Mode	Tunnel Mode
IP Header Protected	No	Yes
Payload Protected	Yes	Yes
New IP Header Added	No	Yes
Typical Use	Host-to-Host	VPN / Gateway-to-Gateway

IPSec Transport Mode

Definition

- In transport mode, IPSec protects the data received from the transport layer before encapsulation into an IP packet.



Characteristics

- Only the payload is encrypted/authenticated.
- Original IP header is not protected.
- IPSec header and trailer are inserted between:
  - IP header
  - Transport-layer payload

Advantages

- Lower overhead
- Faster processing
- Efficient for direct host communication

Limitation

- IP header remains visible to attackers.

Working of Transport Mode

Communication Process

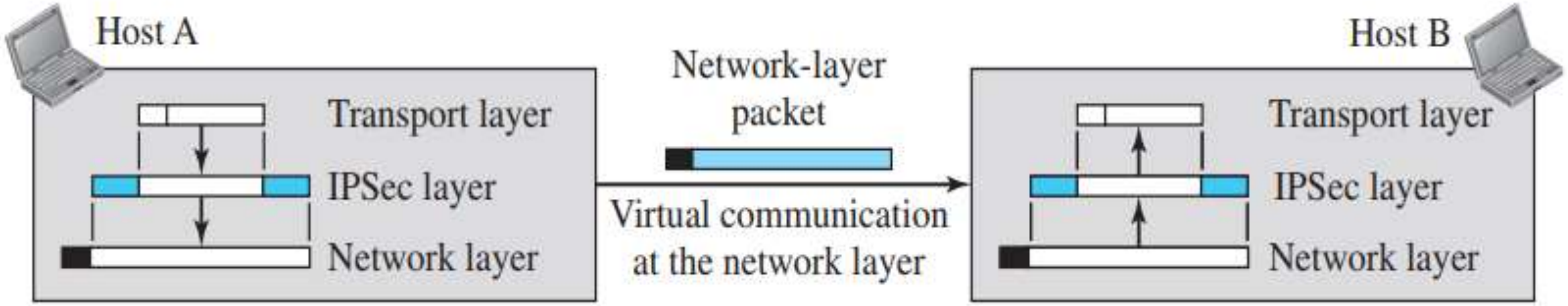
1. Transport layer sends data.
2. IPSec layer encrypts/authenticates payload.
3. IP header is added later by the network layer.
4. Packet travels across the network.
5. Receiver verifies and decrypts payload.

Common Applications

- Secure host-to-host communication
- End-to-end enterprise security
- Secure remote login and communication

Security Provided

- Confidentiality of transport-layer
- Integrity checking
- Sender authentication



PSec Tunnel Mode

Definition

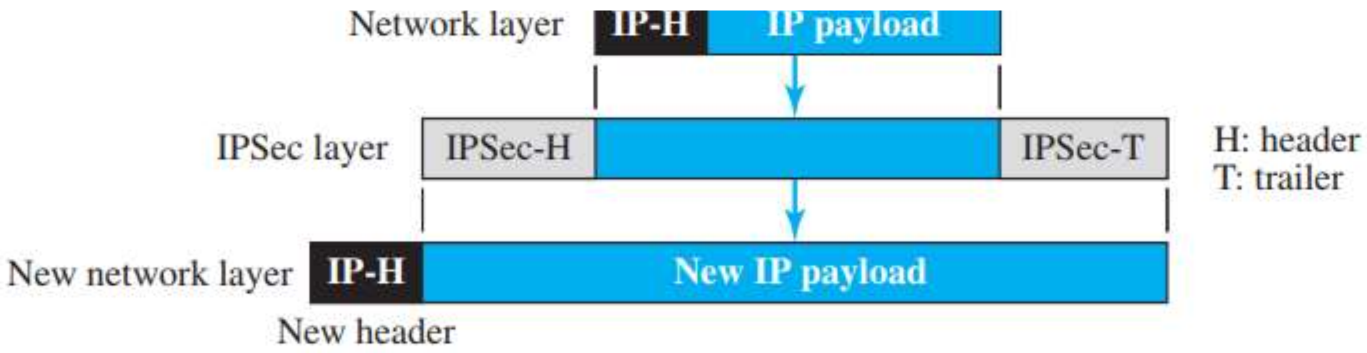
- Tunnel mode protects the entire original IP packet, including the original IP header.

Working Principle

- IPSec encapsulates the complete IP packet.
- A new IP header is attached.
- Original packet becomes the payload of the new packet.

Features

- Hides internal addressing information
- Provides stronger security
- Creates a virtual secure tunnel



Main Use

- VPN implementation
- Secure gateway communication

Working of Tunnel Mode

Communication Flow

1. Original IP packet is created.
2. IPSec encrypts/authenticates the entire packet.
3. New IP header is added.
4. Packet travels securely through the tunnel.
5. Receiving gateway removes IPSec information and restores original packet.

Advantages

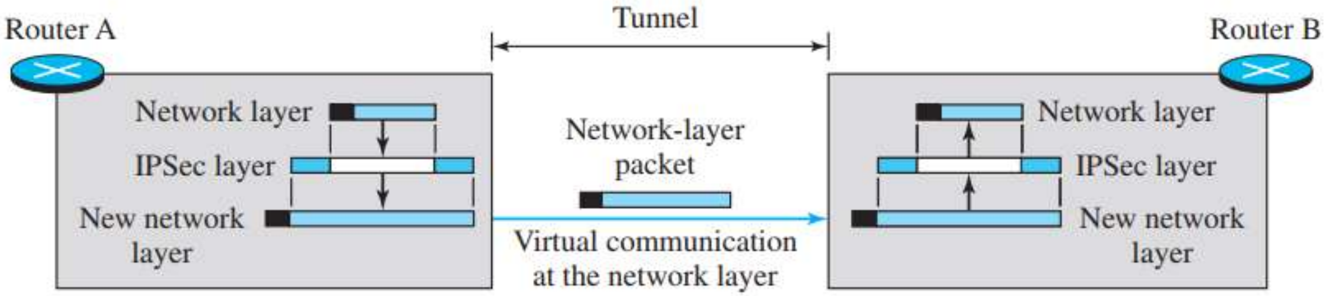
- Full packet protection
- Conceals internal network structure
- Highly secure for Internet communication

Real-World Example

- Corporate VPN connecting branch offices securely over the Internet.

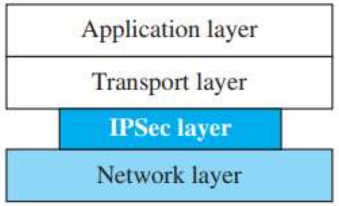
Common Communication Types

- Router-to-router
- Host-to-router
- Router-to-host

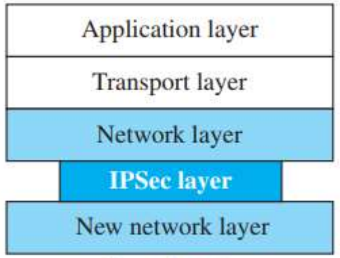


Transport mode versus tunnel mode

Feature	Transport Mode	Tunnel Mode
Protection Scope	Only Payload	Entire IP Packet
Original IP Header	Visible	Hidden
New IP Header	Not Added	Added
Overhead	Low	High
Security Level	Moderate	High
Performance	Faster	Slightly Slower
Common Usage	Host-to-Host	VPNs & Gateways



Transport mode



Tunnel mode

IPSec Security Protocols

IPSec defines two major protocols:

1. Authentication Header (AH)

- Provides:
  - Source authentication
  - Data integrity
- Does NOT provide confidentiality (encryption)

2. Encapsulating Security Payload (ESP)

- Provides:
  - Authentication
  - Integrity
  - Confidentiality through encryption

Purpose of These Protocols

- Protect IP packets against:
  - Tampering
  - Spoofing
  - Replay attacks
  - Unauthorized access

Important Protocol Numbers

Protocol	IP Protocol Number
AH	51
ESP	50

Authentication Header (AH)

Definition

- AH is an IPSec protocol designed to:
  - Authenticate the sender
  - Verify data integrity

Main Characteristics

- Uses:
  - Hash functions
  - Symmetric secret keys
- Generates a message digest for verification.

Security Features

✔ Source Authentication

✔ Data Integrity

✗ No Confidentiality (No Encryption)

AH Placement

- Inserted between:
  - IP Header
  - Payload

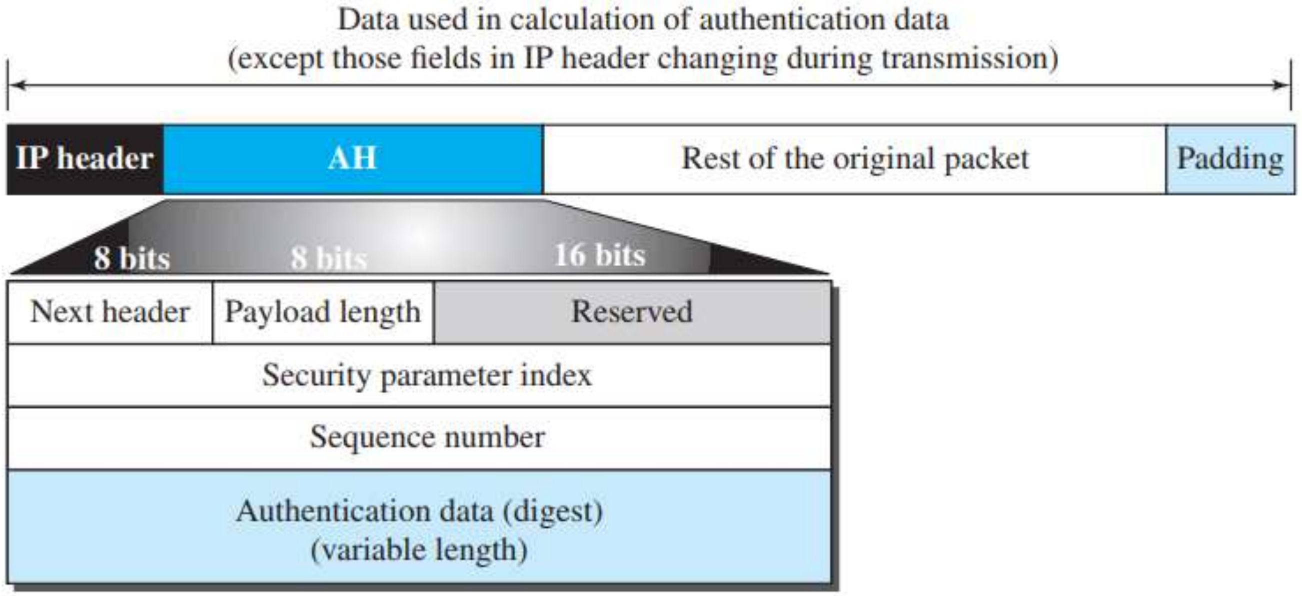
Important Note

- AH protects against packet modification during transmission.

Working of AH Protocol

Steps in AH Processing

1. Authentication header is added.
2. Authentication data field is initialized to 0.
3. Optional padding is added.
4. Hash value is calculated using:
    - Payload
    - Immutable IP header fields
5. Message digest is inserted into AH.
6. IP header protocol field is changed to **51**.



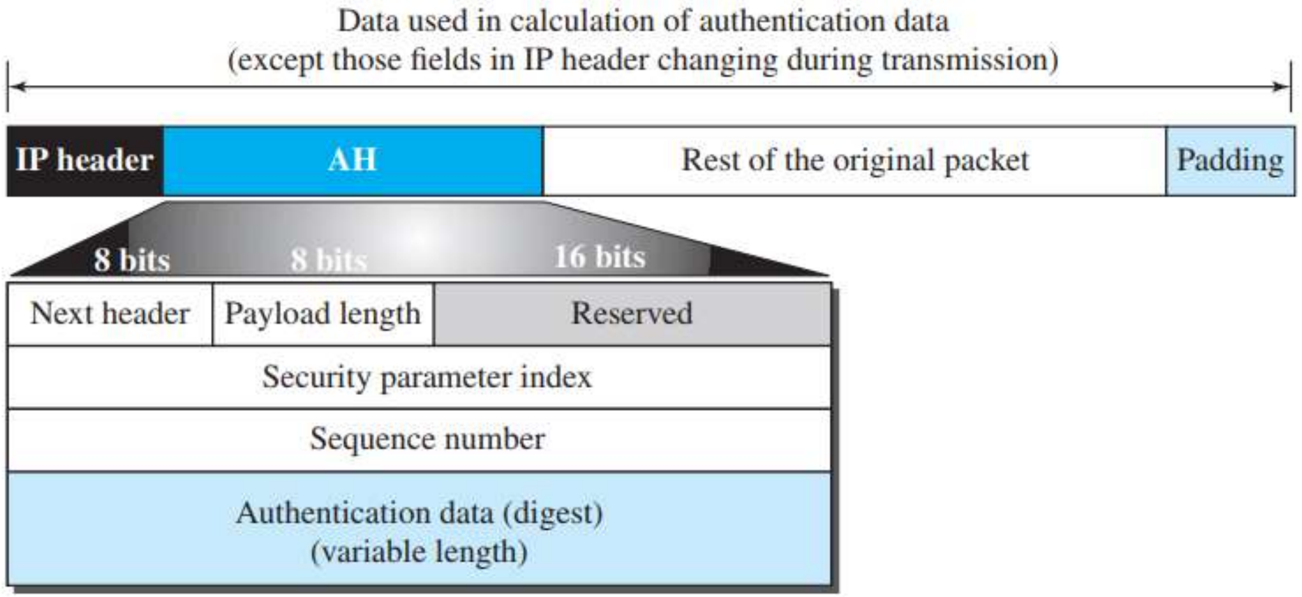
Key Point

- Fields that change during transmission (like TTL) are excluded from hash calculation.

Fields of AH Protocol

1. Next Header

- Identifies the type of payload:
  - TCP
  - UDP
  - ICMP
  - OSPF



2. Payload Length

- Specifies AH length in 4-byte multiples.

3. Security Parameter Index (SPI)

- Identifies the Security Association (SA).

4. Sequence Number

- Prevents replay attacks.
- Provides packet ordering.

5. Authentication Data

- Contains message digest generated using hashing.

Encapsulating Security Payload (ESP)

Definition

- ESP is an IPSec protocol providing:
  - Confidentiality
  - Authentication
  - Data integrity

Main Advantage

- Unlike AH, ESP supports encryption.

Components Added by ESP

- ESP Header
- ESP Trailer
- Authentication Data

Protocol Field Value

- IP protocol number for ESP is **50**.

ESP Protects

- ✔ Payload confidentiality
- ✔ Packet authentication
- ✔ Data integrity

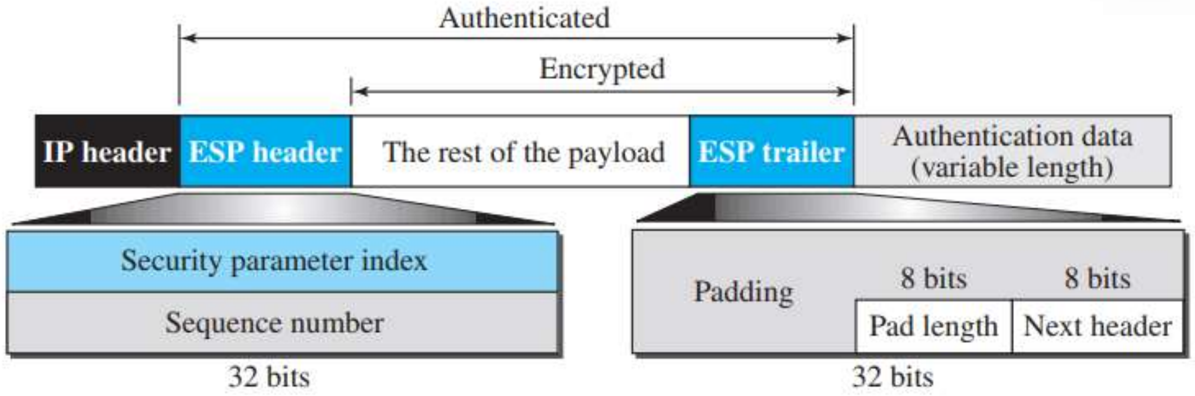
Widely Used In

- VPNs
- Secure Internet communication
- Enterprise networking

Working of ESP Protocol

Steps in ESP Processing

1. ESP trailer is added.
2. Payload and trailer are encrypted.
3. ESP header is inserted.
4. Authentication data is generated.
5. Authentication data is appended at the end.
6. IP protocol field is changed to **50**.



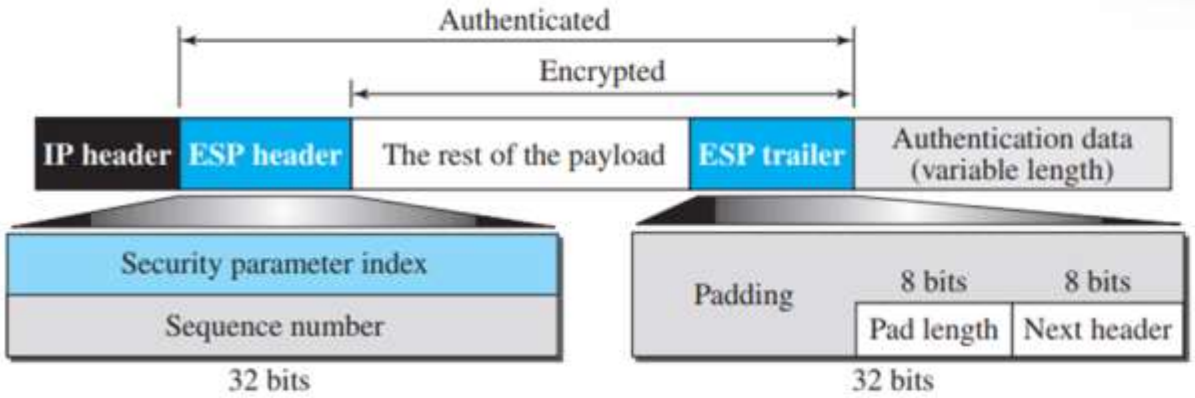
Key Difference from AH

- ESP encrypts data.
- AH only authenticates data.

Benefits

- Strong privacy protection
- Secure data transmission over public networks

Fields of ESP Protocol



1. Security Parameter Index (SPI)

- Identifies Security Association.

2. Sequence Number

- Prevents replay attacks.

3. Padding

- Added for encryption alignment.

4. Pad Length

- Indicates padding size.

5. Next Header

- Identifies encapsulated protocol.

6. Authentication Data

- Used for integrity verification.

Important Feature

- Authentication data in ESP is placed at the end of the packet for easier calculation.

Services Provided by IPSec

Services	*AH*	*ESP*
Access control	Yes	Yes
Message authentication (message integrity)	Yes	Yes
Entity authentication (data source authentication)	Yes	Yes
Confidentiality	No	Yes
Replay attack protection	Yes	Yes

- ESP provides all major security services.
- AH does not provide confidentiality.

Access Control in IPSec

Definition

- IPSec controls access using the:
  - Security Association Database (SAD)

Working Principle

- When a packet arrives:
  - IPSec checks whether a valid Security Association (SA) exist:
- If no valid SA exists:
  - The packet is discarded.

Purpose

- Prevent unauthorized packet transmission.
- Ensure only trusted devices communicate.

Key Benefit

- Adds controlled secure communication at the IP layer.

Message Integrity

- Ensures data is not modified during transmission.
- Sender creates a digest using hashing algorithms.
- Receiver verifies the digest.

Entity Authentication

- Verifies identity of sender.
- Achieved using:
  - Security Associations
  - Keyed hash digest

Supported By

✔ AH
✔ ESP

Result

- Receiver can trust:
  - Data authenticity
  - Sender identity
  - Packet integrity

Confidentiality & Replay Protection

Confidentiality

- Provided only by ESP.
- Uses encryption to hide packet contents.

Important Point

- AH does NOT provide confidentiality.

Replay Attack Protection

- Prevents attackers from resending captured packet
- Uses:
  - Sequence numbers
  - Sliding receiver window

Sequence Number Features

- Starts from 0
- Increases continuously
- Maximum value:

```
2^32 - 1
```

Security Association (SA)

Definition

- A Security Association is a logical secure relationship between two communicating devices.

Purpose

- Converts IP's connectionless communication into a secure connection-oriented service.

SA Stores

- Encryption algorithm
- Authentication algorithm
- Secret keys
- Security parameters

Types of SA

1. Outbound SA
2. Inbound SA

Important Feature

- SA is unidirectional.
- Two SAs are needed for bidirectional communication.

Working of Security Association

Communication Scenario

- Alice communicates securely with Bob.

Security Process

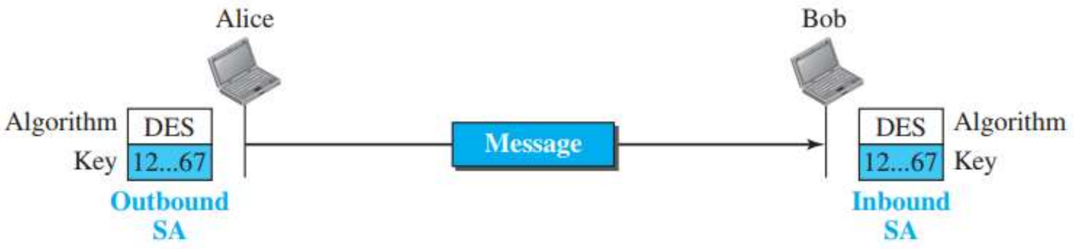
- Shared secret key established.
- Encryption algorithm selected.
- Sender encrypts message.
- Receiver decrypts message using same key and algorithm.

Example Components

Component	Example
Algorithm	DES
Shared Key	Secret symmetric key

Benefits

- Secure communication channel
- Authentication support
- Confidential data transfer



Security Association Database (SAD)

Security Association Database (SAD)

Definition

- SAD stores all Security Associations.

Why SAD is Needed

- Large networks require multiple SAs.
- Supports:
  - Multiple users
  - Multiple destinations
  - Bidirectional communication

Types

1. Inbound SAD
2. Outbound SAD

SAD Entry Contains

- Sequence Number
- Anti-replay window
- IPSec mode
- Security parameters
- Destination address
- Protocol type

Index	SN	OF	ARW	AH/ESP	LT	Mode	MTU
< SPI, DA, P >							
• • •							
< SPI, DA, P >							

Security Association Database

SN: Sequence number
OF: Overflow flag
ARW: Anti-replay window
AH/ESP: Information
LT: Lifetime
Mode: IPSec mode flag
MTU: Path MTU
SPI: Security parameter index
DA: Destination address
P: Protocol